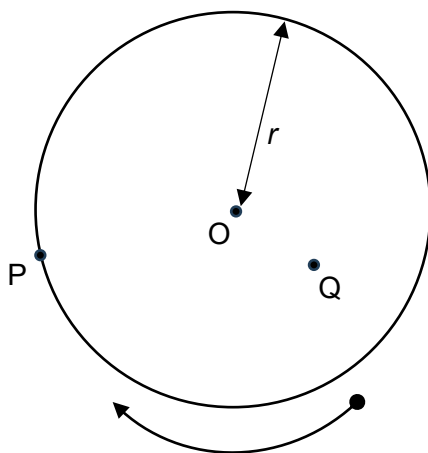


- 25** An aluminium disc of radius r rotates about its centre at a constant speed. It is placed in a uniform magnetic field perpendicular to its surface. A steady electromotive force (e.m.f.) E is generated between the centre O and the rim at P.



What is the e.m.f. generated between points Q and P, where Q is a distance $\frac{r}{2}$ from the centre?

- A** zero **B** $\frac{E}{4}$ **C** $\frac{E}{2}$ **D** $\frac{3E}{4}$

Ans: D

E.m.f. between O and P is $E = B\pi r^2 f$

E.m.f. between O and Q is $E = B\pi \left(\frac{r}{2}\right)^2 f = \frac{E}{4}$

E.m.f. between Q and P is $E - \frac{E}{4} = \frac{3E}{4}$

